

Meal Ingredient Detection: What is on my plate?

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The Problem

Preparing meals often involves many ingredients, but identifying them manually from a photo can be time-consuming and inconsistent. A computer vision system that automatically detects common ingredients can serve as the foundation for applications such as recipe recommendations, meal organization, and nutrition tools.

Solution Overview

This application detects common meal ingredients from a food image using a YOLO object detection model and returns an annotated image with the detected ingredients and confidence scores.



Technical Approach

Component	Selection
Cv Technique	Object Detection
Model	YOLO11
Framework	Ultralytics + PyTorch
Development Environment	Google Colab

YOLO11 was selected for its speed, accuracy, and ability to detect multiple ingredients in a single meal image.

Data Plan

This plan outlines the data source, contents, and preparation steps for the Meal Ingredient Detection system



1. DATASET SOURCE

- Public ingredient detection datasets from Roboflow Universe or Kaggle.
- Example options: Food Ingredients Dataset, Food-101, or custom datasets with ingredient annotations.



2. DATASET CONTENTS

- Annotated meal images containing common ingredients.
- Labels include ingredient names such as tomato, lettuce, chicken, cheese, onion, and more.



3. DATA PREPARATION

- Images will be resized and normalized to match YOLO11 input requirements.
- Dataset will be divided into training, validation, and testing sets.
- Annotations will be formatted in YOLO11 (bounding box) format.



The dataset will provide diverse meal images with clear ingredient annotations to ensure the model learns to detect a variety of ingredient types accurately.

Success Metrics

Metric	Target
Primary Metric	$\text{mAP50} \geq 0.70$
Secondary Metric	Inference time < 1 second per image

The model will be evaluated based on its ability to accurately detect common meal ingredients while maintaining fast inference on Google Colab.

Phase	Goal	Expected Outcome
 Blueprint	Define the project scope and technical approach.	Midterm proposal submitted.
 First Working Demo	Run a pretrained YOLO11 model on sample meal images.	End-to-end pipeline working.
 Make It Yours	Train the model and test ingredient detection.	Ingredient detection system completed.
 Improve & Measure	Evaluate and improve model performance.	Success metrics recorded.
 Package & Present	Final documentation, demo video, and presentation.	Final project submitted.

Milestone Plan

Risks & Resources



POTENTIAL RISKS

RISK	PLAN B
1 A suitable ingredient detection dataset is unavailable.	 Use another public dataset from Roboflow Universe or Kaggle.
2 The model does not achieve the expected performance.	 Reduce the number of ingredient classes and fine-tune the model with a smaller dataset.



RESOURCES

RESOURCE	SELECTION	ESTIMATED COST
 Compute	Google Colab	Free